POLYTECHNIQUE MONTRÉAL

LE GÉNIE EN PREMIÈRE CLASSE

REGISTRARIAT

Produit par Geade le 2026-07-07

RAPPORT D'EXAMINATEUR D'UN MÉMOIRE DE MAÎTRISE OU D'UNE THÈSE DE DOCTORAT

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RENSEIGNEMENTS SUR L'EXAMINATEUR

Instructions : **Rapport à compléter par chaque examinateur, sauf le directeur et le codirecteur de recherche de l'étudiant. À retourner au directeur de département avant la présentation du mémoire ou la soutenance de la thèse. Ce rapport est exigé pour qu'il y ait convocation à la présentation ou à la soutenance. Une copie est transmise au Registrariat pour le dossier de l'étudiant.**

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Signature de l'examinateur

24 septembre 2026
Date

PARTIES 1 - 3

Inscrire vos commentaires au verso ou y joindre votre rapport.
Les aspects ci-dessous devraient être considérés.

PARTIE 1 : VALEUR SCIENTIFIQUE

Originalité du sujet; cohérence dans la structure et l'organisation des parties du mémoire ou de la thèse; utilisation adéquate de la documentation; choix de méthodes appropriées; rigueur dans l'argumentation et le traitement des sources et des données ainsi que dans l'analyse des résultats et dans leur interprétation; acquisitions de la méthodologie de recherche et des connaissances dans le domaine; contribution au développement technologique ou à l'avancement des connaissances.

PARTIE 2 : QUALITÉ DE LA PRÉSENTATION DU DOCUMENT

Précision et clarté du titre et du style; qualité de la présentation et de l'écriture; correction de la langue[1]; qualité des illustrations et des tableaux; choix et présentation des références bibliographiques.

PARTIE 3 : AUTRES COMMENTAIRES

Jugement d'ensemble; commentaires généraux ou spécifiques.

[ ] Selon vous, ce mémoire ou cette thèse devrait-il ou devrait-elle être soumise à un comité d'excellence?

RAPPORT SYNTHÈSE

La présentation du mémoire ou la soutenance de la thèse est recommandée	[X]	Le mémoire ou la thèse est à réviser(corrections majeures)	[ ]	Le mémoire ou la thèse est jugé(e) inacceptable	[ ]

[1] S'il y a lieu, indiquez en annexe la liste des corrections mineures de langue et les erreurs typographiques.

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PRIÈRE DE DACTYLOGRAPHIER (Utiliser des pages supplémentaires si nécessaire)

PARTIE 1 : VALEUR SCIENTIFIQUE

The thesis presents **SparseDiffEngine**, an open-source C library that recasts CVXPY's *matrix stuffing* as structured sparse differentiation: a symbolic pass run once per problem family, a numeric pass repeated at every solve, and two matrix representations (*permuted dense*, *stacked permuted dense*) that keep dense blocks intact and process them with BLAS kernels. The resulting **diffengine** backend is compared with CVXPY's three existing backends on the public benchmark suite: it is the fastest on the large majority of problems for the first canonicalization, and re-solves of parametric problems run at times comparable to CVXPY's cached DPP tensor.

Strengths. The topic is original and relevant: compilation time is a blind spot of the modelling-language community. The structure is exemplary, with a running example carried through every chapter. The literature review is rich and connects three sub-fields (modelling languages, algorithmic differentiation, sparse linear algebra). The experimental methodology is sound: public suite, Dolan–Moré performance profiles, published scripts and timings, and negative results reported honestly. The technological contribution is substantial and real: the backend has been merged into CVXPY (opt-in, parameter-free problems for now) and the student is a co-author of two recent CVXPY preprints.

Reservations. (1) *Positioning.* The *matrix-free* approach of Diamond and Boyd (2016), which answers the same question with the opposite solution (never form the matrix), is neither cited nor discussed; the antecedents of the two core ideas (symbolic/numeric separation in sparse factorizations, block-sparse Jacobians in finite elements or Ceres) are not named, nor are CVXcanon, CVXGEN or the recent rewrite of Convex.jl. The novelty is real but architectural, and must be stated frankly. (2) *Experimental analysis.* The results chapter is thin relative to the rest. No ablation isolates the contribution: the speedup cannot be attributed to the dense representations, to the two-pass split, or simply to a leaner C implementation. This ablation (diffengine with and without the dense representations) is mandatory for the final version; without it the thesis shows *that* the engine is faster but not *why*, which is insufficiently rigorous for a research thesis. In addition, the central claim (caching without DPP) is not tested on any non-DPP problem, each measurement is run once, and instance sizes and profiling are absent. (3) *Delineation.* SparseDiffEngine is also the differentiation engine of the DNLP extension (Cederberg, Zhang, Nobel, Boyd); the thesis must state what pre-existed and what is its own. No complexity analysis, even elementary, is given.

PARTIE 2 : QUALITÉ DE LA PRÉSENTATION DU DOCUMENT

The English is excellent and the pedagogical progression remarkable; Chapters 1 and 2 could serve as teaching material. Figures are careful and useful (2.3, 3.2, 3.6–3.8), tables well designed, bibliography uniform. The title is precise (the engine covers only the back end of canonicalization, which the text states clearly). Minor corrections: “librairie” → “bibliothèque” in the French abstract; “24 of 25” (§4.3) should be qualified as a comparison with cvxcore alone, otherwise it contradicts “21 of 25”; harmonize the designation of the main contribution (§1.5, Ch. 3, §3.5); Figure 3.5 relies on colour alone; “p.” → “pp.” in [41], [42], [43], [76], [88]; references [60], [72], [21] incomplete; update the status of the merged pull request.

PARTIE 3 : AUTRES COMMENTAIRES

Overall judgement. With respect to the aims of a master's thesis at Polytechnique, the exposition of acquired knowledge and the quality of the communication are well above average, and the software realization is substantial and validated by the CVXPY maintainers. The demonstration of research aptitude is, however, incomplete: the experimental chapter does not meet the level of rigour expected of a research thesis. The claimed contribution is not isolated (no ablation), the central claim is not tested on the case that motivates it (no non-DPP problem), measurements are single runs without variance, and neither instance sizes nor a time breakdown are given. The thesis shows *that* the engine is faster, not *why*. These weaknesses are serious enough that I hesitated to request major revisions; I recommend the presentation only on the understanding that the experimental corrections listed below are fully carried out before the final deposit, and I will verify them.

Required before final deposit: (i) add to the literature review a discussion of the *matrix-free* approach and the missing references above, and delineate the thesis's own contribution relative to DNLP; (ii) add an ablation study (diffengine with and without the dense representations) on the full benchmark suite; (iii) add at least one non-DPP parametric problem to the re-solve experiments; (iv) repeat every measurement and report means and standard deviations; (v) report the dimensions of every instance (variables, constraints, nonzeros, parameters); (vi) provide a time breakdown by phase (conversion, symbolic pass, numeric pass, BLAS) that explains the underperforming problems. A detailed report with specific comments and questions for the defence is provided separately.